SPX collars: payoff, pricing and implementation

Omitted: the full annual index series, annual audit tables, historical return/wealth charts, start-month heatmap, downloadable datasets and selected historical regime analysis. The original 19-page report is not cleared or republished by this adaptation. All retained numerical pricing examples are explicitly hypothetical.

This is a newly prepared, text-only public adaptation, not the full original research report. It retains selected explanations and mathematical derivations. Original section numbers and source-reference numbers are retained. Source links appear at the end. Interactive controls and graphics are not reproduced.

01 / How the collar works

Three positions create one expiry payoff

The protection comes from the put; the funding contribution comes from the call sold. The investor also needs underlying equity exposure. For the clean illustration, use matching notionals, the same one-year expiry, a put strike equal to the starting reference level, and a call strike 9% above it.

1

Hold the underlying exposure

Participate in changes in the reference price. In practice this means an equity basket or tracking vehicle; the index itself is a calculation.

2

Buy a protective put

At expiry, the put pays when the reference settles below its strike. Buying this protection costs a premium.

3

Sell a call

Receive a premium and owe the payoff above the call strike. This gives up part of the underlying’s upside.

Mechanics: OIC protective collar; contract details are checked separately below.

[10]

Expiry algebra · normalised reference level starts at 100

The price-related outcome is bounded between 100 and 109

$VT = ST + \max(Kp - ST, 0) - \max(ST - Kc, 0)$

For $Kp = 100$ and $Kc = 109$: $VT = \max(100, \min(ST, 109))$.

At expiry: Reference 70 | Underlying value: 70 | Put payoff: +30 | Short-call payoff: 0 | Combined value: 100 | Price payoff: 0%

At expiry: Reference 105 | Underlying value: 105 | Put payoff: 0 | Short-call payoff: 0 | Combined value: 105 | Price payoff: 5%

At expiry: Reference 120 | Underlying value: 120 | Put payoff: 0 | Short-call payoff: -11 | Combined value: 109 | Price payoff: 9%

Own mathematical illustration; amounts per 100 of reference notional. Matched notionals; same expiry; $Kp \leq Kc$. No option premium or other cash flow included.

[10]

100 at expiry

Underlying 80 + put 20 − call 0 = 100. Price payoff: 0%.

One maturity is different from repeated annual resets.

The historical illustration re-establishes its floor at the start of every calendar year and compounds each year’s transformed return. It is not a single 65-year option or a 9% cap on the whole 65-year gain. Actual reset terms would depend on the market every year.

06 / Can the call pay for the put?

Zero net premium is a market-pricing condition

Choosing a 0% floor and a 9% cap does not make the premiums equal. At a specified timestamp, a conservative separate-leg screen is put ask minus call bid. A package quote may differ; available strikes, size and liquidity still have to support the actual execution.

Net option debit d = Pask(Kp, T) − Cbid(Kc, T).

Zero net option premium requires d = 0; it does not eliminate costs or surrendered upside.

Indicative theoretical example · not a live or historical quote

In this model, the desired 9% cap leaves a debit

Keep the 0% price floor

Put strike = 100

Solve equal model premiums

Call strike ≈ 103.302907

Resulting model cap

≈ 3.302907%, rather than 9%

All inputs are hypothetical: S0 = 100; put strike 100; initial call strike 109; one year; continuously compounded risk-free rate 3%; continuous dividend yield 1.5%; flat implied volatility 20% at both strikes. European Black-Scholes-Merton valuation; no bid-ask spreads, fees, discrete-strike limits, funding charges or skew.

Model outputs: put 7.07706385; call 5.01066477; net debit 2.06639908 per 100 notional. Equal model premiums give a call strike of 103.30290744. Extra digits support reproduction, not quote accuracy.

Independent Black–Scholes–Merton calculation, assumptions set for this illustration on 6 Nov 2026. OIC explains why market premiums, not formulas, determine actual prices.

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[12]

[13]

Capital accounting

The premium changes the return denominator

Paying the model debit requires 102.06639908 of initial capital. If the expiry price component is 100, the return on that expenditure is approximately -2.024564%. At an expiry value of 109, it is approximately +6.793226%.

Return on capital = VT ÷ (100 + d) − 1

This example excludes dividends and all other costs. Borrowing the debit instead creates a separate financing cash flow. A 0% reference-price floor is not a 0% return on a larger initial expenditure.

Independent calculation using the model debit above; indicative illustration only.

[11]

Feasibility condition

Solve for a feasible cap or floor

Fix the put strike and solve for the call strike; or fix the cap and solve for the put strike. A real zero-premium collar is the result of that pricing exercise. It is not the result of clipping a chart.

Strike ordering matters. Retain Kp ≤ Kc. In the stated continuous-yield model, an ATM put cannot be financed by a call at or above spot when the interest rate is below the dividend yield; at equal rates, the equal-premium cap is 0%.

This is a model feasibility condition from put-call parity, not a universal quote rule. Market skew and execution frictions require actual prices.

Independent model algebra; theoretical valuation only.

[11]

What can change the premium balance?

Change, other things equal: Raise the put strike | What it does: More protection; put value rises | Implication for the collar: More call proceeds or a larger debit is needed

Change, other things equal: Raise the call strike | What it does: More upside retained; call value falls | Implication for the collar: Less funding for the put

Change, other things equal: Increase downside implied volatility relative to call volatility | What it does: Put protection becomes relatively dearer | Implication for the collar: The same 0% / 9% pair can cost more

Change, other things equal: Higher rates in the stated model | What it does: Call value rises; put value falls | Implication for the collar: Can improve funding for the same strikes

Change, other things equal: Higher dividend assumptions in the stated model | What it does: Call value falls; put value rises | Implication for the collar: Can worsen funding for the same strikes

Change, other things equal: Change tenor or volatility across both legs | What it does: Both values change | Implication for the collar: Net effect depends on strikes, skew and the joint inputs

OIC option pricing and April 2026 skew discussion; directions for rates and dividends refer to the stated European continuous-yield model.

[11]

[12]

[13]

The historical options backtest remains unsubstantiated.

No synchronized historical put-ask / call-bid archive was obtained. The missing evidence includes reset timestamps, contract availability, strike increments, expiry and settlement values, sizes, execution rules, dividends, funding and costs. Daily collar volatility and drawdown additionally require marks for the actual contracts held.

Instrument history also limits the backtest. Cboe introduced SPX options in 1983. Earlier index years are necessarily counterfactual for listed-SPX execution; any executable backtest needs its own verified start date, available tenors and contract specifications. [18]

Execution and cash flows determine the account return

A clean expiry diagram is a starting point for implementation. Dividend income, actual holdings, option settlement, costs, funding and the investor’s base currency all affect the account result.

Check the exposure and the contract together

Issue: Dividends | What matters in practice: SPX price returns exclude dividends. A basket or ETF may distribute cash; reinvestment changes exposure. Treat those flows and taxes separately. Clipping total returns does not create SPX option payoffs. [4]

Issue: Notional and basis | What matters in practice: Match the option reference, hedge size and dates to the owned exposure. ETF tracking, expenses, portfolio differences and contract-size rounding can leave residual gains or losses. [14]

Issue: Exercise and settlement | What matters in practice: SPX is European-style and cash-settled, with a USD 100 multiplier per index point. Traditional SPX uses AM settlement; SPXW weekly and month-end contracts use PM settlement. [8]

Issue: AM versus closing values | What matters in practice: AM settlement uses constituent opening trades in a special opening quotation. It can differ from an ordinary index close or a displayed value at one instant. Calendar-year returns do not establish the return at an option’s actual settlement. [9]

Issue: Assignment | What matters in practice: European SPX options do not have early assignment. American-style equity or ETF options can, including around dividends; that is a different implementation. [8] [10]

Issue: Cash and collateral | What matters in practice: Cash-settled options do not deliver or remove the shares in the account. Plan for cash obligations. Qualifying ETF coverage may receive margin relief, subject to broker policies and ongoing coverage requirements. [15]

Current Cboe documentation and OCC / OIC education, reviewed 6 Nov 2026. No broker account or product term sheet has been inspected.

[4]

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Timing and costs

A maturity floor is not an early-exit guarantee

Before expiry, both options still have market values. Changes in implied volatility, skew, remaining time, rates and dividends affect those values. Closing the combined position realises available prices, not the expiry diagram.

Liquidity and execution can matter most when protection is needed. Spreads and the ability to close both legs together can change the result; closing one leg leaves a different exposure. Exchange and broker fees, funding, taxes and operational errors also belong in the implementation review.

OCC liquidity, combination and margin risks; OIC pricing inputs. No daily historical collar marks were obtained.

[12]

[14]

Currency exposure

A USD floor is not an SGD floor

Currency conversion is a separate source of return. For an unhedged SGD-based account, let FX be SGD per USD:

$$R_{SGD} = (1 + R_{USD}) \times (FX_{end} \div FX_{start}) - 1$$

Illustrative example: a 0% USD result with a 5% fall in SGD per USD gives -5% in SGD. This is arithmetic, not an FX forecast. Hedging currency introduces its own pricing and cash flows.

Nominal protection also does not preserve purchasing power against inflation. No current FX price is used in this report.

The legal wrapper changes the risk.

A listed-options position, an OTC contract and an issuer note are different arrangements. If exposure is packaged into an OTC or issuer product, counterparty or issuer credit, collateral, liquidity and early-termination provisions must come from its documentation. No actual product terms are inferred here.

Sources and scope

https://www.optionseducation.org/strategies/all-strategies/collar-protective-collar

https://www.optionseducation.org/advancedconcepts/black-scholes-formula

https://www.optionseducation.org/optionsoverview/options-pricing

https://cdn.cboe.com/resources/spx/spx-fact-sheet.pdf

https://www.theocc.com/company-information/documents-and-archives/options-disclosure-document

No current market quotation, historical execution backtest, personalised borrowing amount or guaranteed result is supplied. A public source link is not permission to redistribute its underlying database, screenshots or report.